INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME	

CLASS		INDEX NUMBER	

CHEMISTRY 8872/01

Paper 1 Multiple Choice

18 September 2013

50 minutes

Additional Materials: Data Booklet
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A, B, C,** and **D**. Choose the **one** you consider to be correct.

1 A giant molecule contains a large amount of carbon, mainly of isotopes ^{12}C and ^{13}C. It was found that the relative atomic mass of carbon in the molecule is 12.20.

What is the ratio of ^{12}C and ^{13}C?

A 3:1 **B** 4:1 **C** 3:4 **D** 1:4

2 20.0 cm^3 of 0.02 mol dm^{-3} aqueous sodium bromate(V), $NaBrO_3$, was found to react completely with 80.0 cm^3 of 0.01 mol dm^{-3} hydroxylamine, NH_2OH. The half equation for the reduction of bromate (V) ions is given as shown.

$$BrO_3^-(aq) + 6H^+(aq) + 6e \longrightarrow Br^-(aq) + 3H_2O(l)$$

Which of the following could be the nitrogen containing product in this reaction?

A N_2O

B NO_2

C NO

D NO_3^-

3 The successive ionisation energies (IE) of two elements, **A** and **B**, are given below.

IE/ kJ mol^{-1}	1st	2nd	3rd	4th	5th	6th	7th	8th
A	1090	2350	4610	6220	37800	47000	-	-
B	1251	2298	3822	5158	6542	9362	11018	33604

What is the likely formula of the compound that is formed when **A** reacts with **B**?

A **AB** **B** $\mathbf{A_2B_3}$ **C** $\mathbf{AB_4}$ **D** $\mathbf{A_4B}$

4 Which electronic configuration represents an atom of an element that forms a simple ion with a charge of –3?

A $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$

B $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

C $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$

D $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$

5 Which of the following species has a square planar structure?

A BrF_4^-

B BF_4^-

C CH_4

D C_2H_4

6 Which of the following will not form a hydrogen bond with another of its own molecule?

A CH_3OH

B CH_3CHO

C CH_3NH_2

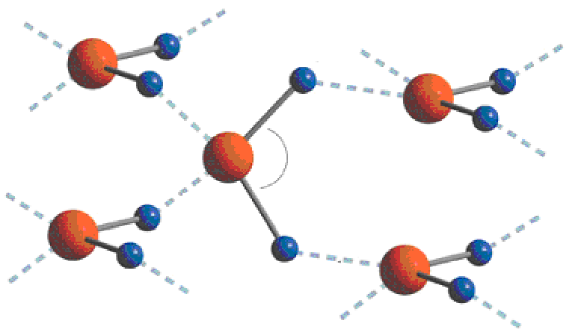
D CH_3COOH

7 The C_2H_2 molecule is linear.

What can be deduced from this about the numbers of σ and π bonds present in the molecule?

	σ	π
A	2	2
B	2	3
C	3	1
D	3	2

8 The diagram below shows part of the structure of ordinary ice.



Which of the following statements is **not true**?

A The bond angle in ice is 104.5º.

B Ice has a cage-like "open" structure.

C The density of water is higher than that of ice.

D Both ice and water have covalent bonding as well as hydrogen bonding.

9 The equilibrium constant for reaction **I** below is K. What is the equilibrium constant for reaction **II**?

reaction **I** $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$

reaction **II** $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

A $2K$

B $2K^{-1}$

C K^{-2}

D K^2

10 A 10 cm^3 sample of 0.010 mol dm^{-3} HC*l* is diluted by adding distilled water at constant temperature.

Which one of the following items correctly shows the effect of the dilution on the concentrations of H^+ and OH^- ions in the solution?

	$[H^+]$	$[OH^-]$
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

11 The table gives the concentrations and pH values of the aqueous solutions of two compounds, **C** and **D**. Either compound could be an acid or a base.

	C	**D**
Concentration	2 mol dm^{-3}	2 mol dm^{-3}
pH	6	9

Student **E** concluded that **C** is a strong acid.
Student **F** concluded that the extent of dissociation is lower in **C**(aq) than in **D**(aq).
Which of the students are correct?

A **E** only

B **F** only

C Both **E** and **F**

D Neither **E** nor **F**

12 The following data refers to the reaction

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$$

$[NO]$/ mol dm^{-3}	$[O_2]$/ mol dm^{-3}	relative initial rate
1.0	0.5	1.0
2.0	0.5	4.0
2.0	1.0	8.0

What is the rate equation for the reaction?

A rate = $k[O_2]$

B rate = $k[NO_2]^2$

C rate = $k[NO][O_2]$

D rate = $k[NO]^2[O_2]$

13 The reaction

$$2\mathbf{G} + \mathbf{H} \longrightarrow \mathbf{J}$$

is first order with respect to **G** and second order with respect to **H**. What is the unit of the rate constant?

A s^{-1}

B mol dm^{-3} s^{-1}

C mol^{-1} dm^{3} s^{-1}

D mol^{-2} dm^{6} s^{-1}

14 Some enthalpy changes of combustion are given below.

	ΔH_c / kJ mol^{-1}
$CO(g) + ½ O_2(g) \longrightarrow CO_2(g)$	–283
$H_2(g) + ½ O_2(g) \longrightarrow H_2O(l)$	–286
$CH_3OH(l) + 1½ O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$	–715

What is the enthalpy change of the following reaction?

$$CO(g) + 2H_2(g) \longrightarrow CH_3OH(l)$$

A –146 kJ mol^{-1}

B –140 kJ mol^{-1}

C +140 kJ mol^{-1}

D +146 kJ mol^{-1}

15 The properties of the oxides of four elements **K**, **L**, **M** and **N** in the Third Period in the Periodic Table are given below.

- The oxide of **K** is insoluble in water and in dilute acid but is soluble in hot and concentrated sodium hydroxide.
- The oxide of **L** is amphoteric.
- The oxide of **M** reacts with dilute sodium hydroxide at room temperature.
- The oxide of **N** dissolves in water to form a strong alkaline solution.

Which of the following is correct in order of increasing proton number?

A **N**, **L**, **K**, **M**

B **K**, **L**, **M**, **N**

C **N**, **K**, **L**, **M**

D **N**, **M**, **K**, **L**

16 The size of Na^{+}, Mg^{2+} and Al^{3+} is in the order: $Na^{+} > Mg^{2+} > Al^{3+}$. Which of the following best explain this trend?

A The number of electrons decreases while the number of protons increases.

B The number of electrons are the same but the number of protons increases.

C The number of electrons and protons decreases.

D The number of electrons and protons increases.

17 Element **O** is heated in chlorine. The product is then added to water. The resulting solution is found to be neutral. What could **O** be?

A sodium

B aluminium

C phosphorus

D chlorine

18 Which pair of reactions could have the same common intermediate?

P $CH_3CH_2CH_3 \longrightarrow$ intermediate $\longrightarrow (CH_3)_2CHCN$

Q $CH_3CH(OH)CH_3 \longrightarrow$ intermediate $\longrightarrow (CH_3)_2C(OH)CN$

R $CH_3CH{=}CH_2 \longrightarrow$ intermediate $\longrightarrow CH_3CH(OH)CH_3$

S $CH_3CO_2CH_2CH_2CH_3 \longrightarrow$ intermediate $\longrightarrow CH_3CH_2CH_2Br$

A **P** and **Q**

B **P** and **R**

C **Q** and **S**

D **R** and **S**

19 High energy radiation in the stratosphere produces free-radicals from chlorofluoroalkanes, commonly known as CFCs.

Which free radical is most likely to result from the irradiation of $CHFClCF_2Cl$?

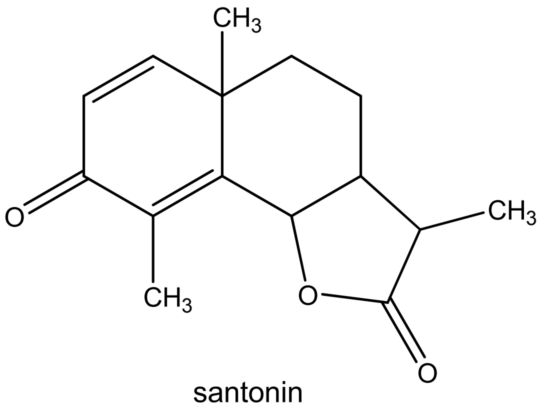
A $CHFCl\dot{C}FCl$

B $\dot{C}HClCF_2Cl$

C $\dot{C}HFCF_2Cl$

D $\dot{C}FClCF_2Cl$

20 Santonin is a drug that was once widely used to expel parasitic worms from the body.



santonin

When santonin is first treated with warm dilute H_2SO_4, and then the product of this reaction is treated with cold acidified $KMnO_4$, a final product **T** is obtained.

How many atoms of hydrogen in each molecule of product **T** can be displaced with sodium metal?

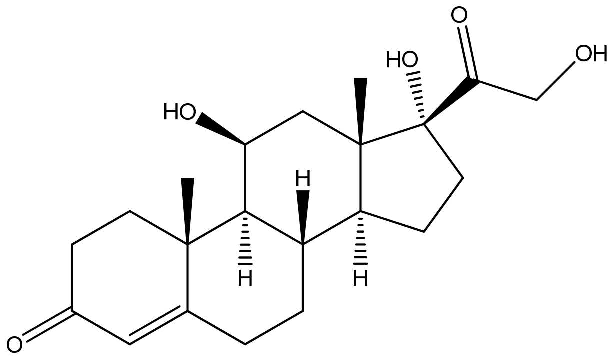
A 2

B 4

C 5

D 6

21 Hydrocortisone is a steroid hormone produced by the adrenal gland and is released in response to stress. It is commonly used as an active ingredient in anti-inflammatory creams.



Hydrocortisone

Which of the following statements about hydrocortisone is incorrect?

A It does not show geometric isomerism.

B It turns hot acidified $K_2Cr_2O_7$ from orange to green.

C When treated with an excess of hot concentrated acidified $KMnO_4$, it forms a compound containing five carbonyl groups.

D When treated with $NaBH_4$ in the presence of methanol, it forms a compound containing five hydroxy groups.

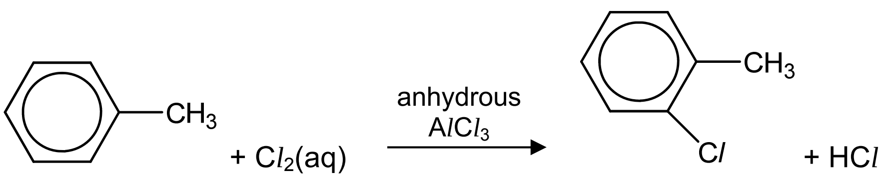
22 The reaction conditions for four different transformations are given below.

Which transformation has a set of conditions that is **not** correct?

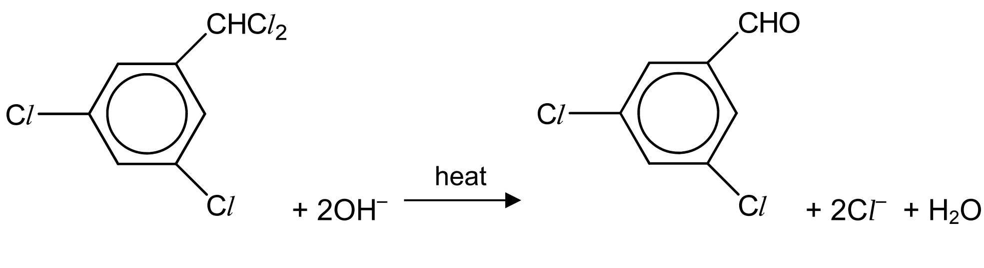
A $CH_3CH_3 + Cl_2\ (CCl_4) \xrightarrow{\text{UV light}} CH_3CH_2Cl + HCl$

B $CH_2{=}CHCH{=}CH_2 \xrightarrow[\text{Heat}]{KMnO_4/H^+} 4CO_2 + 3H_2O$

C



D



23 An alcohol with molecular formula $C_nH_{2n+1}OH$ does not react with MnO_4^- / H^+.

What is the least number of carbon atoms such an alcohol could possess?

A 4

B 5

C 6

D 7

24 An organic compound **U** has the following properties.

- changes the colour of acidified sodium dichromate(VI) from orange to green,
- has no effect on Fehling's reagent,
- gives a positive tri-iodomethane test

What could **U** be?

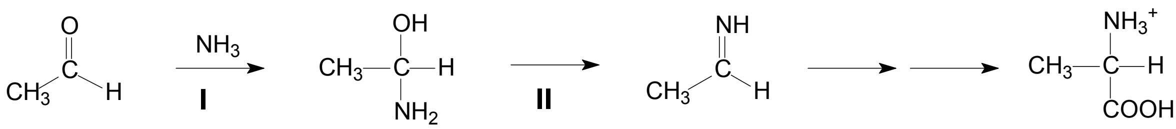
A CH_3COCH_2CHO

B $CH_3COCH_2COCH_3$

C $CH_3COCH_2CH_2OH$

D $HOCH_2CH_2CH_2CH_2OH$

25 The following reaction scheme shows the synthesis of an amino acid, alanine.



alanine

Which of the following shows the correct types of reaction for steps **I** and **II**?

	I	II
A	substitution	elimination
B	substitution	condensation
C	addition	elimination
D	addition	addition

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

26 Which of the following ions contains half-filled d-orbitals?

1 Mn^{2+}

2 Cu^{+}

3 Fe^{2+}

27 Which of the following processes is endothermic?

1 electrolysis of water

2 freezing of water

3 condensation of steam

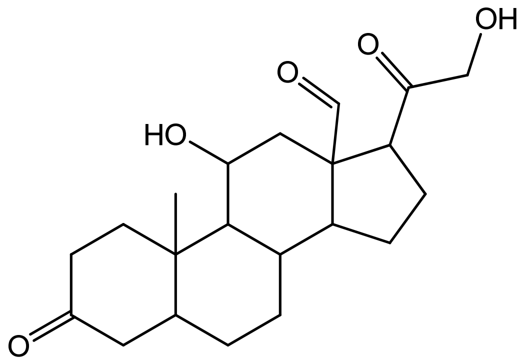
28 Which of the following might alter the rate constant of a reaction?

1 temperature

2 concentration

3 pressure

29 Aldosterone is a steroid hormone that increases reabsorption of ions and water in the kidney. It has the structure shown below.



Aldosterone was reacted in separate experiments with:

(i) 2,4-dinitrophenylhydrazine
(ii) $Ag(NH_3)_2^+(aq)$
(iii) $PCl_5(s)$

Which statements about these reactions are correct?

1 One mole of aldosterone could react with three moles of 2,4-dinitrophenylhydrazine.

2 One mole of aldosterone could form two moles of Ag.

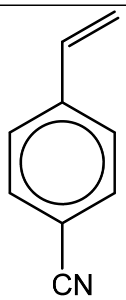
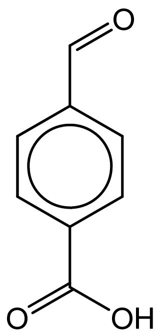
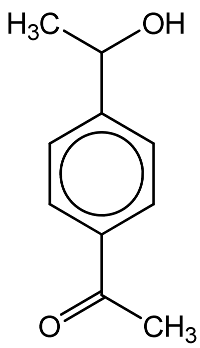
3 One mole of aldosterone could react with 2 moles of $PCl_5(s)$.

The responses **A** to **D** should be selected on the basis of

A	**B**	**C**	**D**
1, 2 and **3** are correct	**1** and **2** only are correct	**2** and **3** only are correct	**1** only is correct

No other combination of statements is used as a correct response.

30 Which of the following reactions would produce 1,4-dicarboxylic acid?

	reactant	reagents and conditions
1		hot acidified potassium manganate(VII)
2		hot acidified potassium dichromate(VI)
3		alkaline aqueous iodine and heat followed by acidification